

Intro

Phishing attacks have been present in the world since the 1990s when people would phish AOL to acquire credentials granting free wifi [1], [2]. Phishing has grown in popularity due to the fact that trying to breach a company's firewall which protects user data is much more difficult than targeting the user who has the sensitive data [1]. Our team is entering the space of machine learning based phishing-URL detection which has a vast literature of existing solutions [3], [4], [5] [6]. The new contribution we're seeking to make is in understanding which threat factors and strategies generative AI uses to generate URLs which beat phishing detection models trained on non-AI generated URLs.

The first documented case of fraud was 3400 years ago in 300 BC [7]. A shipping merchant named Hegestratos took out an insurance policy with his co-conspirator Zenothemis against his ships, purposely sank them, and was promptly caught [7]. Fraud is as old as time, and the internet has created a plethora of opportunities for fraudulent behaviour. The first documented instances of internet phishing were in 1990 when malicious actors were changing their AOL display names to those of moderators and then phishing for their login credentials to gain free internet access [1]. In the year 2000, an email with the subject letter ILOVEYOU was sent that infiltrated Windows machines with a worm virus that ended up propagating to an estimated 45 million PCs around the world, showing the potential vulnerability of email messages [8]. In 2001, spoofed websites started being created of popular websites like eBay and PayPal in an effort to compromise manipulated users' credentials [2]. \$929M was lost in 2004-2005 from phishing campaigns [2]. In 2025, the FBI reported that in the United States of America alone losses from phishing totalled \$215.8M [9].

Most phishing campaigns rely on social engineering, the method of attacking a user through manipulative means to convince them to share sensitive information under a false pretext. As a result, phishing email detection is a large space in the cyber security realm. Companies use network level protections (blacklists and anti-spam techniques), authentication, content analysis, website blacklists and whitelists, similarity based detection, and feature-based detection (link features, structural features, and word list features) [10]. Advances in generative AI are allowing people to create fraudulent attacks at unprecedented pace, rendering many prior solutions incompetent for various reasons [11]. Some detection methods are too computationally expensive to use (authentication, content analysis, feature based structural detection and word lists) and others cannot keep pace with the speed that AI generated fraudulent websites are being created (whitelists and blacklists). Research in 2008 showed that the average phishing URL had a lifespan of only 3 days [12], and while there isn't existing research on the topic of generative AI fraudulent website lifespans, it can be assumed that they're much faster to create. Phishing detection through URL analysis on the other hand is one of the most cost effective methods of preventing phishing, as not many characters need to be analyzed to determine the validity of a phishing attempt [5]. Many existing URL detection methods use machine learning models trained on existing datasets of phishing URLs to detect URLs through common features like IP addresses being present, common misspellings, and subdomains of seemingly legitimate domains [10]. It's been proven that these models can be easily beaten by models like DeepPhish which are trained to create URLs that beat existing detection methods [3]. To our

knowledge there is not any existing literature looking at how traditional machine learning URL phishing detection models perform against URLs created by generative AI and the threat factors that generative AI use to beat historical models.

Sources

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